Semantic Architecture for Sustainable Disposal: A Comprehensive Data Discovery Report for the "Fessi" Knowledge Graph

1. Introduction: The Semantic Gap in Waste Management Systems

The development of "Fessi," a knowledge graph designed to power a waste disposal chatbot for a German university, represents a significant challenge in semantic engineering. The fundamental difficulty in automating waste disposal guidance lies not in the regulations themselves, which are codified and static, but in the semantic gap between the user's vocabulary and the regulatory framework. University staff and students interact with the world through a product-centric lens—purchasing "toner cartridges," "coffee cups," or "lab centrifuges." Conversely, waste management regulations, specifically the German Waste Catalogue Ordinance (*Abfallverzeichnis-Verordnung* - AVV), operate through a material-centric and hazard-centric lens, classifying items as "waste printing toner," "composite packaging," or "electrical equipment containing hazardous components."

To bridge this gap, the knowledge graph must function as a robust translation layer, capable of ingesting diverse product taxonomies and mapping them to appropriate waste streams. This report details the discovery, analysis, and architectural integration of universal taxonomies required to populate Fessi. It addresses the user's specific requirement for standardized classification systems—including UNSPSC, GPC, ECLASS, HS Codes, CPC, and NACE/WZ—and identifies machine-readable data sources (XML, JSON, CSV) that allow for the automated ingestion of these hierarchies. Furthermore, it explores the critical role of "grey literature" sources, such as e-procurement catalogs and open-source repositories, in supplementing official government datasets.

The transition from a product's useful life to its end-of-life status requires a deep understanding of its provenance. By integrating procurement data (what the university buys) with waste data (how it must be discarded), Fessi can infer disposal instructions with high precision. For instance, an item identified via its United Nations Standard Products and Services Code (UNSPSC) as a "Lithium Polymer Battery" inherits chemical attributes that automatically trigger a mapping to AVV hazardous waste codes, removing the cognitive burden of technical classification from the user. The following sections provide an exhaustive technical roadmap for acquiring and structuring this data.

2. The Procurement-Waste Nexus: United Nations Standard Products and Services Code (UNSPSC)

The United Nations Standard Products and Services Code (UNSPSC) is a hierarchical classification system used extensively in global commerce and procurement. Within the context of a German university, UNSPSC is particularly valuable because it often underlies the financial Enterprise Resource Planning (ERP) systems (e.g., SAP, Oracle, Kuali) used for purchasing equipment and supplies. Ingesting this taxonomy allows Fessi to recognize items based on the university's own procurement ledger, creating a direct link between the "Buy" signal and the "Dispose" signal.

2.1 Taxonomy Architecture and Granularity

The UNSPSC employs a rigorous four-level hierarchy codified into an eight-digit number, with an optional fifth level for business function. Understanding this structure is critical for designing the node depth of the Fessi knowledge graph.

- **Segment (Digits 1-2):** This represents the logical aggregation of families for analytical purposes. For example, *Segment 44* corresponds to "Office Equipment and Accessories and Supplies".[1] In the graph, this high-level node serves as a primary filter, allowing the system to categorize broad clusters of waste (e.g., "Office Waste").
- **Family (Digits 3-4):** A commonly recognized group of inter-related commodity categories. *Family 12* within Segment 44 (*4412*) denotes "Office Supplies".[1]
- **Class (Digits 5-6):** A group of commodities sharing a common use or function. *Class 17* (*441217*) identifies "Writing Instruments".[1] This level often correlates with the material composition groups found in municipal waste instructions (e.g., rigid plastics, wood, metal).
- **Commodity (Digits 7-8):** The specific good or service. *Commodity 06* (*44121706*) specifies "Wooden Pencils".[1] This is the leaf node where specific disposal attributes (e.g., "biodegradable," "sharps") can be attached.

The optional fifth level, "Business Function," adds two digits (making a 10-digit code) to describe the *use* case, such as retail or rental.[2] For waste management, the 8-digit commodity code is the primary target for data ingestion. The granularity provided by the commodity level enables precise material mapping; distinguishing between "Correction Fluid" (solvent-based, hazardous) and "Wooden Pencils" (residual waste) occurs at this specific tier.

2.2 Data Source Discovery and Ingestion Strategy

Accessing the UNSPSC codeset in a machine-readable format is the first step in population. While the United Nations Development Programme (UNDP) and GS1 US manage the standard, "official" access often requires paid membership for the absolute latest version. However, for the purpose of a waste knowledge graph, slightly older versions are perfectly adequate and

are available through open government data portals and academic repositories.

Official Open Government Repositories

The most reliable source for a license-free, bulk-downloadable dataset is the U.S. General Services Administration (GSA) via Data.gov.

- **Source:** Data.gov / State of Oklahoma Office of Management and Enterprise Services.[3]
- **Dataset Title:** "UNSPSC Codes".
- **Format:** The dataset is provided in **CSV** format, which is ideal for batch processing and graph injection.
- **Metadata Quality:** The dataset was updated as recently as November 22, 2024, ensuring that the codes are current enough to include modern electronics and consumables relevant to university operations.[3]
- **Ingestion Logic:** The CSV structure typically includes columns for Segment, Family, Class, and Commodity codes alongside their textual descriptions. The Fessi ETL (Extract, Transform, Load) pipeline should treat this CSV as the "skeleton" of the product ontology.

Academic and Institutional "Grey" Sources

University procurement departments often publish "Cheat Sheets" or subset lists of UNSPSC codes to help staff classify their purchases. These documents are highly valuable for Fessi because they act as a *relevance filter*.

- **Source:** Cornell University Division of Financial Affairs.[1]
- **Format:** Microsoft Excel (.xlsx).
- **Content:** These files list the most commonly purchased commodities in a university setting. Instead of populating the graph with irrelevant codes for industrial mining or heavy weaponry, the system can prioritize the items found in these academic cheat sheets, ensuring the chatbot is optimized for the campus environment.[1]
- **Implication:** By cross-referencing the Data.gov master list with university-specific subsets, the graph can assign a relevance_score to nodes, allowing the search algorithm to prioritize "Lab Beaker" over "Industrial Vat" when a user queries "glass container."

Open Source and Machine Learning Repositories

The developer community on GitHub has produced several repositories containing pre-parsed and "cleaned" UNSPSC datasets, often formatted for use in machine learning models. These sources are excellent for training the chatbot's Natural Language Understanding (NLU) component.

- **Repository:** KathiravanNatarajan/UNSPSCPrediction and kasivisu4/cordiance-experential-project.[5]
- **Files:** UNSPSCdataset.csv, final_result.json.
- **Utility:** These datasets often map natural language descriptions (e.g., "sticky notes") to formal UNSPSC titles ("Adhesive note paper"). This synonym mapping is crucial for the

chatbot to understand colloquial user queries. The JSON formats allow for direct document storage or graph property enrichment.[6]

- **O*NET Resource Center:** The U.S. Department of Labor's O*NET database includes a UNSPSC Reference.xlsx file.[7] This file is particularly structured, separating the code and title for each hierarchy level into distinct columns, simplifying the programmatic reconstruction of the taxonomy tree.

2.3 Semantic Integration and Insights

The integration of UNSPSC into Fessi serves a dual purpose: identification and hazard inference. By mapping high-level UNSPSC Segments to AVV chapters, the system can automate broad categorizations. For example, Segment *12* (Chemicals) can be semantically linked to AVV Chapter *16* (Wastes not otherwise specified in the list) or Chapter *06* (Wastes from inorganic chemical processes).

Table 1 illustrates the proposed mapping schema between UNSPSC hierarchy levels and Fessi Graph Nodes.

UNSPSC Level	**Graph Node Label**	**Property Example**	**AVV Mapping Potential**
Segment	ProductCategory	id: 44000000, label: "Office Equipment"	Low (Too broad)
Family	ProductFamily	id: 44120000, label: "Office Supplies"	Medium (General Waste)
Class	ProductClass	id: 44121700, label: "Writing Instruments"	High (Material inference)
Commodity	ProductItem	id: 44121706, label: "Wooden Pencils"	Exact (Specific AVV Code)

The data suggests that while UNSPSC provides the "Product" identity, it lacks the "Material" identity. Therefore, UNSPSC nodes must be enriched with material properties derived from other taxonomies, specifically GPC and ECLASS, to fully solve the waste sorting problem.

3. The Consumer Interface: Global Product

Classification (GPC)

While UNSPSC dominates the back-office procurement world, the **Global Product Classification (GPC)** system reigns in the consumer-facing front office. GPC is the classification standard underpinning the GS1 system, which manages the Global Trade Item Numbers (GTINs)—the barcodes found on virtually every retail product sold on campus, from vending machine snacks to bookstore stationery. For a chatbot designed to assist students, the ability to process a barcode scan or a GTIN input is a critical feature, and GPC provides the taxonomy to interpret that data.

3.1 Structural Dynamics: Bricks and Attributes

GPC operates on a four-tier hierarchy similar to UNSPSC but introduces a critical fifth dimension: **Attributes**. This structure is uniquely suited for waste management because it explicitly separates the "what" (the product) from the "how" (the features/materials).

- **Segment:** The highest level, e.g., *Food, Beverage and Tobacco*.
- **Family:** A broad grouping, e.g., *Milk/Butter/Cream/Yogurt/Cheese/Eggs/Substitutes*.
- **Class:** A narrower grouping, e.g., *Milk/Milk Substitutes*.
- **Brick:** The primary unit of classification. A product is assigned exactly one Brick. For example, *Brick 10000025* represents "Milk (Perishable)".[8]
- **Attribute:** This is the game-changer for Fessi. Attributes describe the variants of a Brick. For the "Milk" Brick, attributes might include *Fat Content*, *Organic Claim*, and critically, *Packaging Type*.

In a waste disposal context, the "Brick" tells the graph the *content* (Milk), but the "Attribute" tells the graph the *container* (Carton vs. Glass). This distinction allows Fessi to generate dynamic decision trees. If a student asks about "Milk," the chatbot can query the GPC schema to find associated packaging attributes and ask, "Is it a carton or a glass bottle?" based on the specific BrickAttribute relationships defined in the standard.

3.2 Acquiring the XML Schemas

GS1 makes the GPC schemas available for implementation, though access often requires navigating their "GPC Browser" tool or partner portals.

- **Official GPC Browser:** The primary method for discovery is the GS1 GPC Browser. This tool allows users to browse the hierarchy and, crucially, download the published schemas.[9]
- **File Formats:** The schemas are typically distributed as **XML** files or **Excel (XLSX)** spreadsheets contained within ZIP archives. The XML format is preferred for graph ingestion as it explicitly defines the parent-child relationships and the linkage between Bricks and valid Attributes.[8]
- **Version Control:** GS1 publishes updates regularly (e.g., "GPC as of November 2025"). The snippet evidence confirms availability of JSON and XML downloads directly from the

browser interface.[10] The "Delta" documents, which track changes between versions, are also available.[11] Ingesting these Delta files is essential for maintaining the knowledge graph over time, ensuring that new product categories (e.g., "Vapes" or "Plant-based meat alternatives") are recognized as they enter the market.

- **Licensing and Intellectual Property:** It is important to note that GS1 standards are protected by intellectual property policies. While the *structure* can be downloaded for implementation (which includes building internal systems like Fessi), the *redistribution* of the raw data might be restricted. The report recommends using the schemas to build the internal logic of the graph while ensuring compliance with GS1's terms of use.[8]

3.3 Linking GPC to Waste Streams

The integration of GPC into Fessi requires mapping "Bricks" to material types.

- **Scenario:** A student scans a barcode on a plastic yogurt cup.
- **Data Flow:**
  1. User inputs GTIN.
  2. System resolves GTIN to GPC Brick *10000025* (Yogurt).
  3. System retrieves GPC Attribute *Packaging Material: Plastic*.
  4. Graph query: MATCH (m:Material {name: 'Plastic'})-->(b:Bin {type: 'Yellow Sack'}).
  5. Result: "Please dispose of the yogurt cup in the Yellow Bin (Gelber Sack)."

This logic demonstrates why GPC is superior to UNSPSC for consumer goods: the explicit "Attribute" layer provides the material data that UNSPSC lacks.

4. The Industrial Standard: ECLASS

While UNSPSC covers general procurement and GPC covers retail, **ECLASS** is the dominant standard for the European industrial and engineering sectors. Given the university's location in Germany, ECLASS is likely the standard used for procuring technical equipment, laboratory supplies, and furniture. It is the only standard among those analyzed that includes a dedicated structure for "Material Declarations," making it uniquely capable of handling hazardous waste assessment.

4.1 Taxonomy and Material Declaration

ECLASS uses a four-level hierarchy (Subject area, Main group, Group, Sub-group/Product class) encoded in 8 digits, similar to UNSPSC.

- **Example:** *27-01-01-01* corresponds to *Electrical engineering -> Power supply -> Battery -> Lithium battery*.
- **Release 16.0:** The latest version (Release 16.0) has significantly expanded its "material declaration" capabilities. It now includes comprehensive information on critical ingredients to meet regulatory requirements (e.g., REACH, SCIP).[12] This "Block 0173-1#01-AKA421#001" (Declaration of substances) allows specific chemical

compounds to be associated with a product class.

For Fessi, this means the graph can automatically flag items as "Sondermüll" (Hazardous Waste) based on their ECLASS categorization. If a research lab disposes of a component classified under an ECLASS code linked to "Mercury compounds," the chatbot can issue an immediate warning.

4.2 Sourcing ECLASS Data: The "Grey" Advantage

ECLASS is a paid standard, and obtaining the full official release usually requires a license fee. However, for a university project, there are legitimate "grey" pathways to access the necessary data structures without purchasing a commercial license for the entire database.

Amazon Business Punchout Catalogs

A major finding of this research is that **Amazon Business** utilizes ECLASS for its B2B "Punchout" integrations in Germany. To facilitate this, Amazon publishes downloadable taxonomy files for system administrators.

- **Source:** Amazon Business "System Integrations" help pages (DE marketplace).[13]
- **Content:** The snippets identify downloadable lists (likely CSV or Excel) of ECLASS codes that map Amazon's catalog to the ECLASS standard.[13]
- **Strategic Value:** These files represent a curated subset of ECLASS codes that are actively used in commerce. By ingesting these lists, Fessi effectively captures the ECLASS codes for everything available on Amazon Business—which likely covers a vast majority of standard university supplies. This is a high-quality, publicly accessible data source that serves as a practical proxy for the full standard.

Academic and Demo Access

- **Asset Administration Shell (XML):** ECLASS offers a free download of the "Asset Administration Shell" submodels (Asset.xml) to encourage adoption. These XML files provide the structural schema of the standard, allowing Fessi developers to design the graph ontology to be ECLASS-compliant even before populating it with the full code list.[12]
- **University Libraries:** It is highly probable that a German university has institutional access to the full ECLASS database for research purposes. The report advises checking the library's database subscriptions or contacting the engineering faculty.

4.3 Semantic Mapping and REACH

ECLASS codes should be mapped to the **REACH** regulation (Registration, Evaluation, Authorisation and Restriction of Chemicals) within the graph.

- **Graph Model:** (Product:ECLASS_Item)-->(Chemical:Substance)-->(Regulation:REACH).
- **Inference:** If a substance is restricted under REACH, the disposal instruction must default to "Hazardous Waste Collection." ECLASS provides the data linkage to make this

inference automated.

5. The Material Proxy: Harmonized System (HS) and Combined Nomenclature (CN)

The **Harmonized System (HS)**, managed by the World Customs Organization, and its European extension, the **Combined Nomenclature (CN)**, managed by Eurostat, are primarily designed for customs and trade statistics. However, they are invaluable for waste management because customs duties are often determined by *material composition*. An HS code distinguishes between "Articles of leather" and "Articles of plastic" to apply different tariffs. This feature makes HS/CN codes an excellent proxy for material identification in the knowledge graph.

5.1 Structure: The Physics of Trade

- **HS (6 Digits):** Global standard.
- **CN (8 Digits):** EU specific extension.
- **Logic:** The hierarchy is based on material progression.
  - *Section I:* Live Animals.
  - *Section VII:* Plastics (Chapter 39) and Rubber (Chapter 40).
  - *Section XV:* Base Metals (Chapters 72-83).
  - *Section XVI:* Machinery and Electronics (Chapters 84-85).[14]

For Fessi, Section XVI is particularly relevant for identifying **E-Waste** (WEEE). Items classified in Chapters 84 and 85 almost universally map to AVV Chapter 16 (*Wastes not otherwise specified in the list*) or 20 01 36 (*discarded electrical and electronic equipment*).

5.2 Data Source Discovery: Eurostat and TARIC

EU Vocabularies (The New RAMON)

Eurostat has migrated its classification data from the legacy RAMON server to the "EU Vocabularies" platform.

- **Source:** Publications Office of the European Union / EU Vocabularies.[15]
- **Dataset:** "Combined Nomenclature 2024" (and upcoming 2025).
- **Formats:** The data is available as **Linked Open Data (LOD)** in formats like **RDF**, **XML**, and **SKOS**. This is the "gold standard" for a knowledge graph, as the RDF format is native to graph databases.[16]
- **Bulk Download:** XML distributions of CN 2021-2025 are available on the EU Open Data Portal (data.europa.eu). These files allow for the complete reconstruction of the 8-digit hierarchy.[17]

TARIC: The Integrated Tariff

The **TARIC** database integrates the CN codes with specific EU legal measures, such as bans, duties, and restrictions.

- **Data Scale:** The TARIC database is massive. Snippets describe loading a "5 GiB XML file" into XML-native databases like BaseX or eXist-db.[19]
- **Complexity:** Parsing TARIC requires robust XML processing. The data is stored in a complex schema where measures (e.g., "Import Prohibition") are linked to nomenclature codes.[20]
- **Waste Relevance:** TARIC flags specific codes with "Prohibitions and Restrictions" (e.g., Ozone Depleting Substances). If Fessi identifies a product with a TARIC code linked to such a restriction, it effectively identifies a hazardous waste stream without needing a separate chemical database.
- **German Context (EZT-online):** The *Elektronischer Zolltarif* (EZT) is the German implementation of TARIC. While primarily a web interface, the underlying data structure mirrors the EU TARIC XMLs. For a German university, ensuring the graph reflects EZT-specific national codes (11-digit level) adds a layer of precision for compliance.[21]

6. The Missing Link: Central Product Classification (CPC)

While the user request highlighted **CPC (Central Product Classification)**, it is often overshadowed by HS and UNSPSC. However, CPC is the critical statistical bridge between the *production* side (NACE/WZ) and the *trade* side (HS). Managed by the UN Statistics Division, CPC classifies *all* goods and services.

6.1 Role in Fessi

CPC Version 2.1 groups products based on their physical properties and intrinsic nature, much like HS, but with a structure that aligns better with economic output.

- **Bridging Function:** CPC serves as the "Rosetta Stone" between the activity of the university (Education/Research - NACE) and the products it consumes (HS/UNSPSC).
- **Services:** Unlike HS, which only covers goods, CPC covers *services*. This allows Fessi to handle "waste" concepts that are actually services, such as "Septic Tank Cleaning Services" (CPC 94), mapping them to appropriate waste water codes.

6.2 Data Sourcing

- **Source:** UN Stats / Eurostat RAMON (archives) or EU Vocabularies.
- **Format:** Correspondence tables (Excel/CSV) mapping CPC to HS and CPC to NACE are the primary value. These tables allow the graph to traverse from a Service (NACE) -> Product Category (CPC) -> Trade Code (HS).[23]

7. Contextual Filtering: NACE and WZ 2008

To properly categorize waste, one must often know the *source* (the waste generator). A solvent bottle generated by the "Art Department" might be treated differently than one from the "Chemical Engineering Department."

7.1 WZ 2008: The German NACE

The **Klassifikation der Wirtschaftszweige, Ausgabe 2008 (WZ 2008)** is the German national implementation of the EU's NACE Rev. 2.

- **Source:** Destatis (Federal Statistical Office).[24]
- **Format:** Downloadable **Excel** and **XML** files.
- **Implementation:** Fessi should map university departments to WZ codes.
  - *Department of Chemistry* -> **WZ 72.11** (Research and experimental development on biotechnology).
  - *University Canteen* -> **WZ 56.29** (Other food service activities).
- **Benefit:** The AVV waste catalogue defines waste by *chapters* based on source industry (e.g., *Chapter 01: Wastes from exploration...*, *Chapter 20: Municipal wastes*). By knowing the WZ code of the user's department, Fessi can filter the AVV to show only the relevant chapters, reducing the search space and error rate.

8. The Target Ontology: European Waste Catalogue (AVV)

All product taxonomies in Fessi must ultimately resolve to the **Abfallverzeichnis-Verordnung (AVV)** (European Waste Catalogue). This is the legal destination.

8.1 Structure and Logic

- **Chapters (2-digit):** Source-based (e.g., *03 Wood processing*, *20 Municipal waste*).
- **Sub-chapters (4-digit):** Process-based.
- **Codes (6-digit):** The unique identifier (e.g., *20 01 01 Paper and Cardboard*).
- **Hazard Marker:** Codes followed by an asterisk (*) denote hazardous waste (*Sondermüll*).

8.2 Data Sources for Graph Population

- **Official XML:** The German Federal Ministry of Justice (*Gesetze-im-Internet*) publishes the full AVV text in **XML**. This is the authoritative source for the legal German definitions and should be used to populate the label and description properties of the waste nodes.[26]
- **Umweltbundesamt (UBA):** The German Environment Agency provides PDF/CSV downloads of waste statistics which include the code lists.[27]
- **GitHub JSONs:** Developers have parsed the AVV into machine-friendly JSON formats.

    - **Repo:** bennokr/6d5f1e55... contains "European Waste Catalogue (EWC)" as newline-delimited JSON.[28]
    - **Repo:** MarvinJWendt provides a list of German waste terminology (*Abfallverzeichnis*).[29]
    - **Repo:** spherity/product-pass references EWC codes in the context of digital product passports, offering a schema for linking products to waste codes.[30]
- **Recommendation:** Use the GitHub JSONs for the initial graph structure (Nodes/Edges) and the official XML to validate the text and ensuring legal compliance.

9. Semantic Integration: Schema.org and Wikidata

To transform Fessi from a database into a *knowledge graph*, the data must be semantically linked using standard ontologies.

9.1 Schema.org Modeling

- **Product:** Use schema:Product to store UNSPSC/ECLASS/GPC data.
    - **Properties:** sku (internal ID), gtin (barcode), category (taxonomy code), material (linked to a Material node).[31]
- **Waste Management:** Schema.org lacks a dedicated "Waste" type. The best practice is to extend schema:Product or use schema:CreativeWork (for the regulation text). The property schema:usageInfo can be used to link a product to its disposal instructions.[32]

9.2 Wikidata Alignment

Wikidata serves as a massive semantic hub.

- **Property P279 (Subclass of):** Fundamental for hierarchy. *Waste paper* (Q...) is a subclass of *Waste* (Q...).
- **Property P31 (Instance of):** *Banana peel* is an instance of *Biodegradable waste*.
- **P Codes for Waste:** Snippets indicate specific properties for "recycling code" and "European Waste Catalogue code" exist in Wikidata.[33]
- **Querying:** A SPARQL query can be run against Wikidata to retrieve all items that have an EWC code property. This can bootstrap the graph with thousands of pre-mapped product-to-waste relationships.
    - *Query:* SELECT?item?ewc WHERE {?item wdt:P...?ewc }.

10. Data Ingestion Architecture & Architecture

To populate "Fessi," a robust ETL (Extract, Transform, Load) pipeline is required.

10.1 Extraction

1. **UNSPSC:** Download CSV from **Data.gov**.
2. **ECLASS:** Scrape/Download CSVs from **Amazon Business (DE)** help pages.

3. **CN 2025:** Download XML from **EU Vocabularies**.
4. **AVV:** Clone JSON from **GitHub** (bennokr).
5. **GPC:** Download XML from **GS1 GPC Browser**.

10.2 Transformation (Ontology Mapping)

The core challenge is creating the "Golden Record" or "Super-Node" for a concept.

- **Concept:** "Toner Cartridge"
- **Properties:**
  - UNSPSC_ID: 44103103
  - ECLASS_ID: 19-14-06-01
  - CN_ID: 8443 99 90
  - Material:
- **Edges:**
  - (:Concept)-->(:UNSPSC_Node)
  - (:Concept)-->(:AVV_Node {code: '08 03 18'})

10.3 Cross-Reference Tables

Use the **Correspondence Tables** from Eurostat (RAMON/EU Vocabularies) to build the edges between taxonomies.[23]

- **CN to CPA to NACE:** This chain allows the graph to infer that a product (CN) is typical of a specific industry (NACE).

10.4 Graph Schema Design

- **Nodes:** Product, Material, WasteStream, Regulation, TaxonomyTerm.
- **Edges:** IS_A (Hierarchy), COMPOSED_OF (Material), DISPOSE_IN (Bin), REGULATED_BY (Law).

11. Conclusion

The construction of the "Fessi" knowledge graph requires a synthesis of open government data and industry standards. No single taxonomy is sufficient: UNSPSC provides the procurement context, GPC handles consumer barcodes, ECLASS offers the material detail for hazardous waste, and HS/CN connects to global trade definitions. By leveraging the specific machine-readable sources identified in this report—particularly the Data.gov CSVs, Amazon's ECLASS lists, and Eurostat's XML distributions—the university can build a highly intelligent system. This system will not merely list waste codes but will understand the *nature* of products, enabling accurate, context-aware disposal advice that evolves with both the university's supply chain and the regulatory landscape.

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